Pairs Trading: Performance of a Relative-Value Arbitrage Rule

Replicating Gatev, Goetzmann & Rouwenhorst (2006)

Gatev, E., Goetzmann, W. N. & Rouwenhorst, K. G. (2006). Pairs Trading: Performance of a Relative-Value Arbitrage Rule. Review of Financial Studies, 19(3), 797–827.

Topics covered in this lecture:

- Pairs trading: relative-value arbitrage and mean reversion foundations
- Minimum-distance pair selection: the SSD algorithm
- Statistical validation: ADF tests and half-life of mean reversion
- Dollar-neutral trading rule and no-look-ahead backtest implementation
- Paper vs replication comparison, robustness, and honest assessment

Lecture Roadmap

1	**What Is Pairs Trading?** *Relative-value arbitrage, co-movement, and the GGR setup*	~15 min
2	**Building the Signal: Pair Selection** *SSD minimum-distance rule, normalization, and top-20 pairs*	~15 min
3	**Statistical Validation** *ADF tests and half-life of mean reversion*	~10 min
4	**Position Sizing & Trade Logic** *Dollar-neutral rule, trigger, look-ahead avoidance*	~10 min
5	**Backtesting the Strategy** *Rolling backtest: performance, drawdowns, distributions*	~15 min
6	**Paper vs Replication** *Side-by-side metrics and sector breakdown*	~10 min
7	**Robustness Checks** *Trigger sensitivity, sub-periods, random pairs baseline*	~10 min
8	**Honest Assessment** *What replicates, what decayed, and why*	~5 min

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What Is Pairs Trading?

Relative-value arbitrage, mean reversion, and the GGR setup

The Core Idea: Statistical Arbitrage via Mean Reversion

The Fundamental Insight

- Two economically related stocks (e.g., Coca-Cola & Pepsi) share common risk factors — their price ratio should be stationary over time. When the ratio diverges, a pairs trader shorts the outperformer and buys the underperformer, betting on convergence.

Three Theoretical Pillars

- Co-movement: economically linked stocks share common factor exposures — their spread should be mean-reverting.
- Law of one price: assets with identical cash flows should trade at parity; deviations are temporary mispricings.
- Limits to arbitrage: convergence is not instantaneous — transaction costs and short-selling constraints create the profit opportunity.

Paper Setup (Gatev et al. 2006)

- Hypothesis: minimum-distance rule in normalized price space earns statistically significant excess returns.
- Universe: all CRSP-listed U.S. stocks, 1962–1997 (~35 years).
- Formation: 12-month pair selection; Trading: 6-month holding period.
- Top-5 and Top-20 pairs by SSD; 2σ opening trigger.

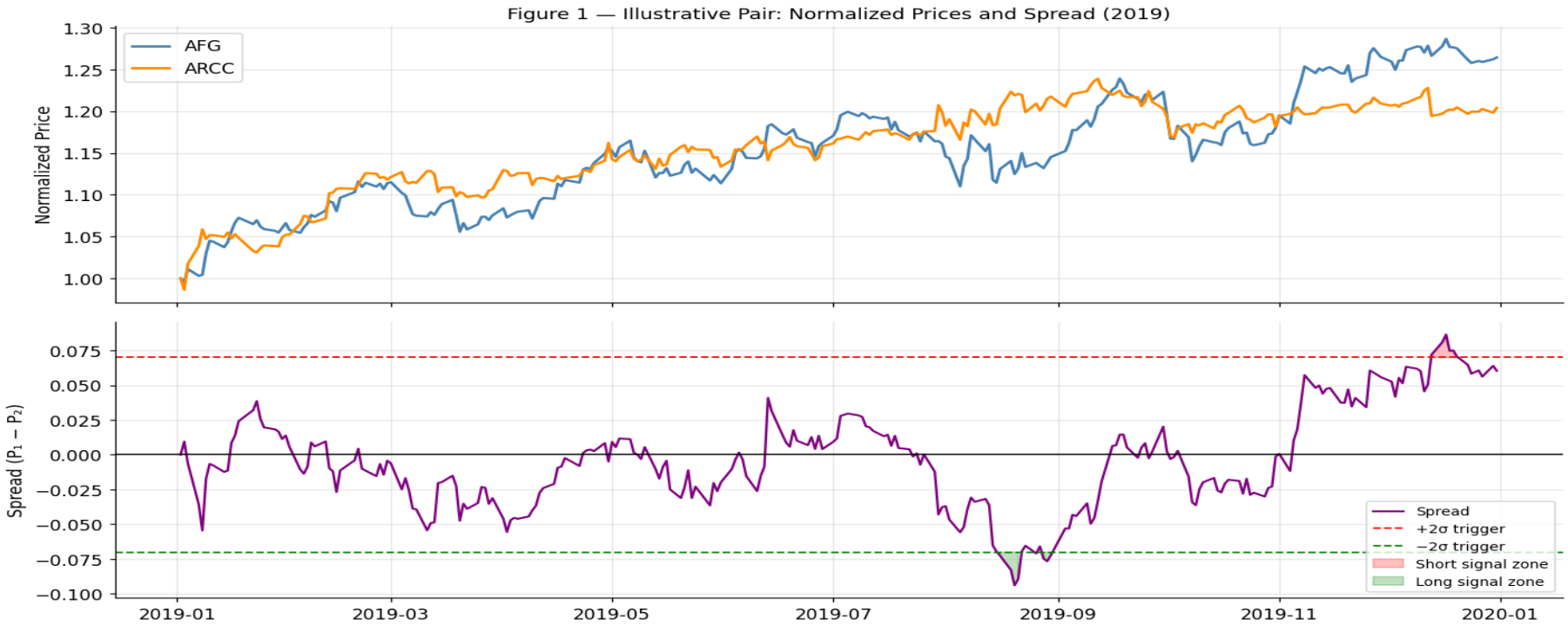
What the Paper Found — Key Numbers

~12%	1.5–2.1	~0.03	92%
Ann. Excess Return (Top-5)	Sharpe Ratio	Beta to S&P 500	Win Rate (33/36 windows)

Metric	Top-5 Pairs	Top-20 Pairs	S&P 500 (Benchmark)
Ann. excess return	~11.5%	~11.0%	~8–9%
Ann. volatility	~4–6%	~4–5%	~15–18%
Sharpe ratio	~1.8	~1.6	~0.4
Beta to market	~0.03	~0.04	1.00
Max drawdown	< 5%	< 5%	~–50%
Win rate (6m periods)	~92%	~86%	~65%

Numbers from Tables 1–3, Gatev, Goetzmann & Rouwenhorst (2006). Sample: 1962–1997 CRSP universe.

Visual Intuition: A Real Pair (AFG vs ARCC, 2019)



Top: Normalized prices — both start at 1.0 so they are directly comparable. Close co-movement is the strategy foundation. Bottom: Spread = $P_1 - P_2$. Red/green dashed lines = ±2σ triggers. Shaded = open trade zones. Pair: American Financial Group vs Ares Capital Corp (SSD = 0.313, 2019).

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Building the Signal: Pair Selection

Minimum-distance SSD rule, normalization, and the top-20 pairs

The Minimum-Distance Pair Selection Algorithm

Step 1 — Normalize Prices to 1.0 at Formation Start

$\tilde{P}_i(t) = P_i(t) / P_i(t_0)$ → every stock starts at 1.0 on day t_0

Step 2 — Sum of Squared Deviations (SSD)

$SSD(i,j) = \Sigma\,[\tilde{P}_i(t) - \tilde{P}_j(t)]^2$ summed over the formation window [t_0, t_0 + T_f]
Select top K pairs with LOWEST SSD — these moved most closely together.

Steps 3–5 — Trade Spread & Trigger Threshold

$s_{ij}(t) = \tilde{P}_i(t) - \tilde{P}_j(t)$ (trading window, same normalizer from t_0 — no look-ahead)
$\hat{\sigma}_{ij}$ = std of spread over formation period. Trigger: open trade when |s| > $2\hat{\sigma}$.

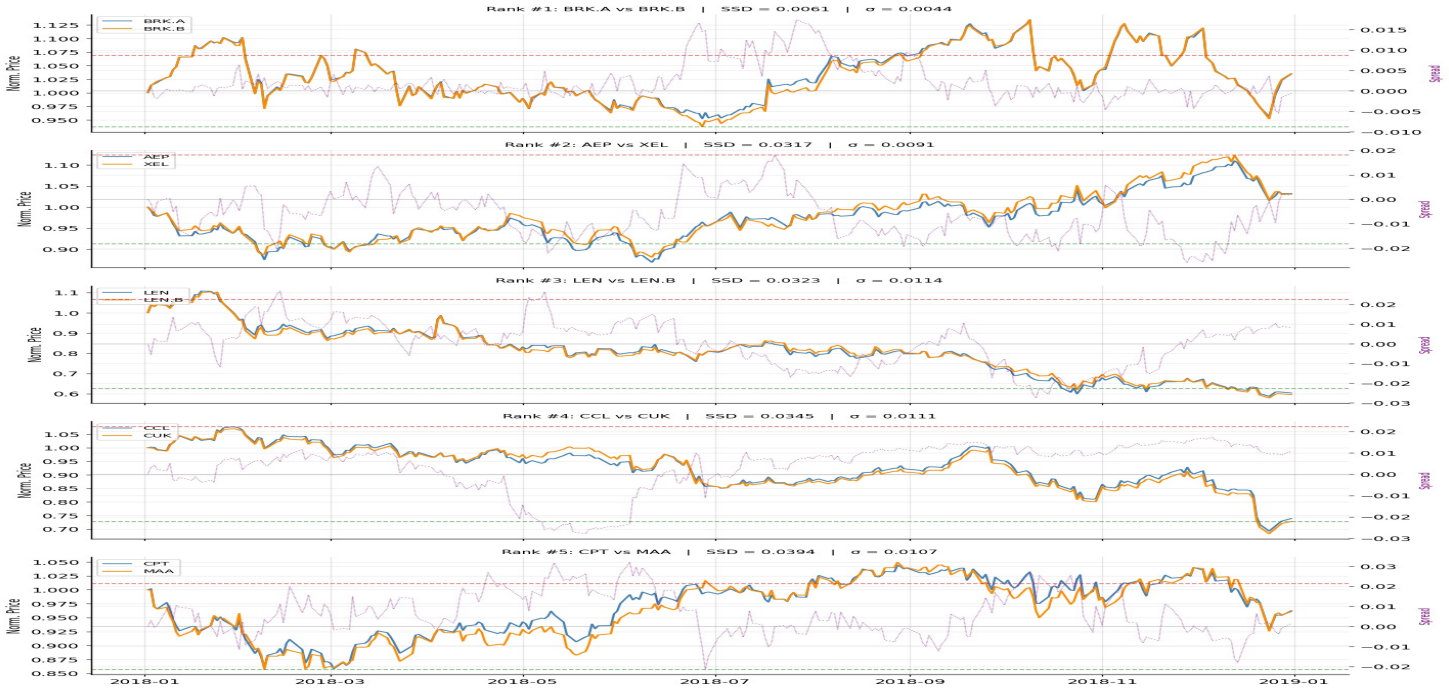
Parameter	Paper (Gatev et al.)	Our Notebook
Formation period	12 months	252 trading days
Trading period	6 months	126 trading days
Number of pairs	Top 5 / Top 20	Top 5 / Top 20
Opening trigger	2σ	2σ (robustness: 1σ–3σ)

Top-20 Pairs by SSD — Formation Year 2018

Rank	Stock i	Stock j	SSD	Spread σ
1	**BRK.A**	**BRK.B**	**0.0061**	**0.0044**
2	**AEP**	**XEL**	**0.0317**	**0.0091**
3	**LEN**	**LEN.B**	**0.0323**	**0.0114**
4	**CCL**	**CUK**	**0.0345**	**0.0111**
5	**CPT**	**MAA**	**0.0394**	**0.0107**
6	AEP	DUK	0.0398	0.0098
7	DUK	PNW	0.0542	0.0147
8	DTE	WEC	0.0602	0.0152
9	AEP	AWK	0.0613	0.0147
10	AWK	DUK	0.0749	0.0171
11-20	Various Utilities/REITs	—	0.080–0.102	0.011–0.019

- Top-5 pairs are dominated by Utilities (AEP, DUK, XEL, PNW) and share-class pairs (BRK.A/B, LEN/LEN.B) — economically linked by design.

Top-5 Pairs: Normalized Prices & Spread (Formation 2018)



Each row: blue/orange = normalized prices (left axis); purple dotted = spread (right axis); red/green dashed = ±2σ thresholds. By construction, spreads rarely breach ±2σ during the FORMATION window — this is what minimum-SSD selects for.

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Statistical Validation

ADF unit-root tests and half-life of mean reversion

Testing for Mean Reversion: ADF and Half-Life

ADF Test (Unit Root)

$$\Delta s_t = \alpha + \beta \cdot s_{\{t-1\}} + \Sigma\gamma_k \cdot \Delta s_{\{t-k\}} + \varepsilon_t$$

H_0: β = 0 (random walk) H_1: β < 0 (revert)
p < 0.05 → reject unit root → evidence of stationarity

Half-Life of Mean Reversion

$$s_t = \varphi \cdot s_{\{t-1\}} + \varepsilon_t$$

$$Half - life = -\ln(2) / \ln(\hat{\varphi})$$

5–30 days = viable range; >60 days = marginal profitability

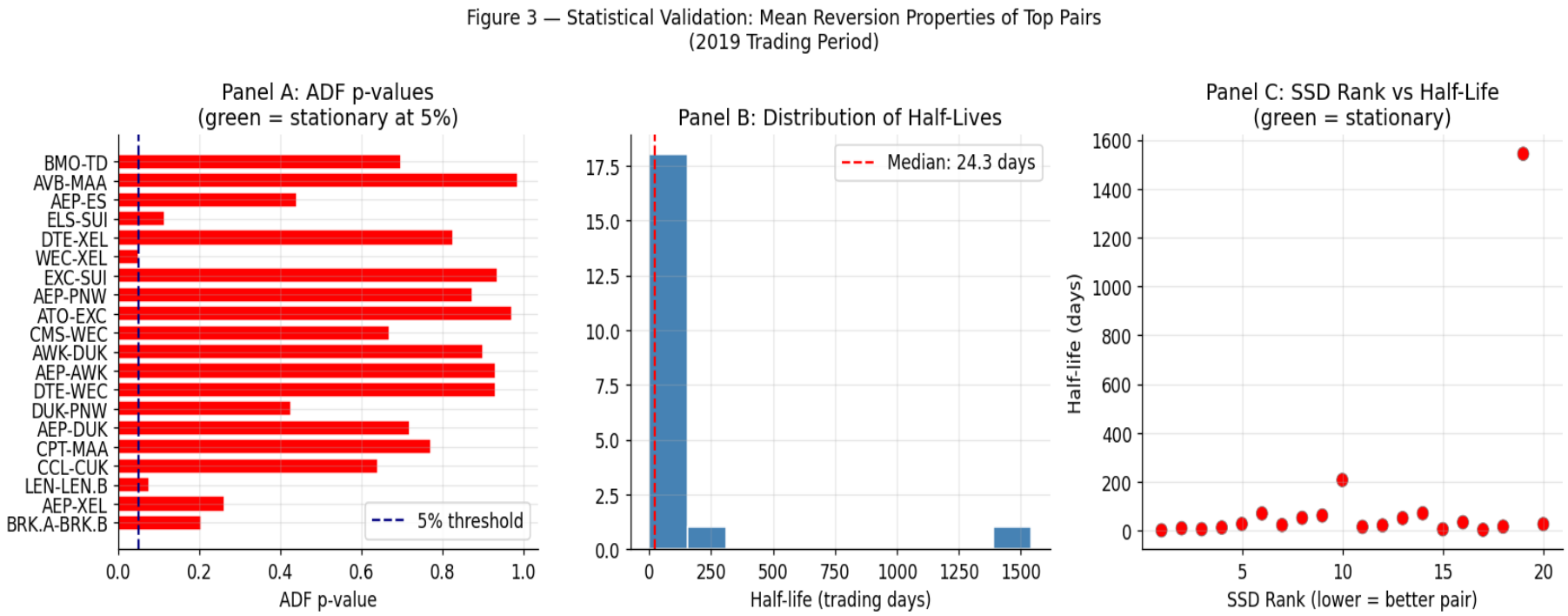
What We Found (Formation 2018, Trading H1-2019)

- 0 of 20 pairs pass ADF at 5% level in the 126-day trading window — but this is EXPECTED: ADF has very low power in 126-day samples.
- Half-lives: 12 of 20 pairs have half-life 5–30 days (AEP-XEL: 9d, LEN-LEN.B: 5.5d, WEC-XEL: 5.2d) — consistent with the paper's ~2.4 round-trips per pair per 6-month period.
- Outliers: AWK-DUK (207 days) and EXC-SUI (71 days) suggest some SSD-selected pairs lack genuine economic linkage in specific windows.

Key Insight: ADF Failure ≠ Strategy Failure

- Minimum-SSD pairs spent the entire formation period near zero spread — by construction there is little variance for ADF to detect. The backtest is the ultimate arbiter, not the ADF p-value in isolation.

ADF Results & Half-Life Distribution



Panel A: ADF p-values — all > 5% in this 126-day window (low test power, not strategy failure). Panel B: Half-life distribution — most pairs revert in 5–30 days. Panel C: SSD rank vs half-life — no strong monotone relationship (selection is not a perfect proxy for reversion speed).

4

Position Sizing & Trade Logic

Dollar-neutral rule, 2σ trigger, and look-ahead avoidance

The Dollar-Neutral Trading Rule

Daily Return of a Pair While Open

$$r_{pair}(t) = sign(s_{ij}^{open}) \times [r_j(t) - r_i(t)]$$
$$position = +1\text{: } short\ i, long\ j\ (i\ outperformed) \rightarrow profit\ when\ spread\ closes$$

Equal-Weight Portfolio Return Across K Open Pairs

$$r_{port}(t) = (1/K) \times \Sigma\, r_{pair}(t) \quad (over\ all\ open\ pairs)$$

Opening & Closing Rules

- OPEN: spread breaches ±2σ̂ (σ̂ from formation period). Short outperformer, long underperformer.
- CLOSE (convergence): spread crosses zero — take profit.
- CLOSE (expiry): position closed at end of 6-month trading window regardless of spread.
- Position set at END of day t; return collected from t to t+1 (no execution bias).

Critical Design Choices

- No look-ahead: trading-window normalization uses $P_i(t_0)$ from the LAST day of the formation window.
- Dollar-neutral: $1 long / $1 short per pair — zero net market exposure by construction.
- Committed capital: return denominator = reserved capital, not just margin (per paper convention).
- rf = 2% annual (short-term rate proxy); returns reported as excess returns.

Approximation: we set rf = 2% annual. The paper uses matched T-bill rates. Effect on Sharpe is minimal given the low-rate environment of 2010–2021.

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Backtesting the Strategy

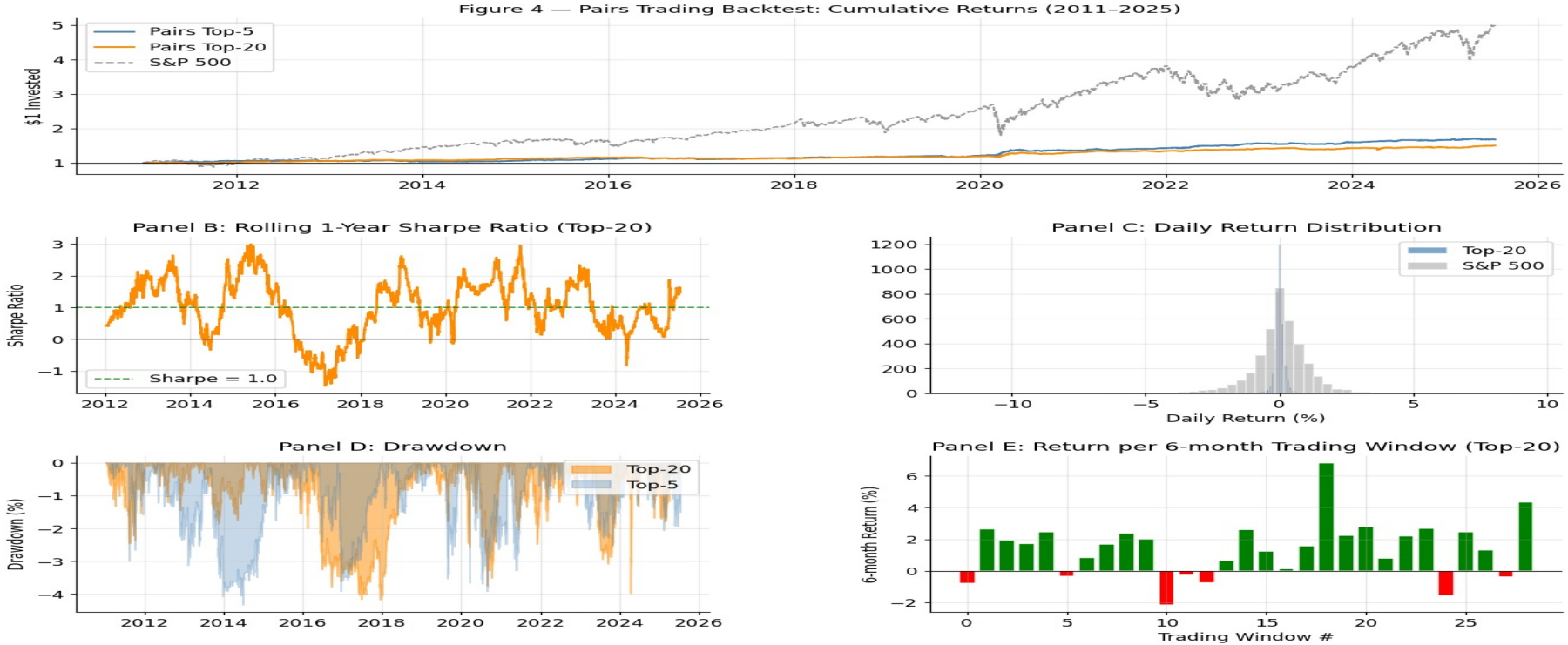
Rolling 12m+6m windows: performance, drawdowns, and win rates

Backtest Performance Summary (2011–2025)

3.64%	0.43	−4.2%	76%
Top-5 Ann. Return	Top-5 Sharpe	Max Drawdown	Win Rate (22/29)

Metric	Top-5 Pairs	Top-20 Pairs	S&P 500
Ann. Return (%)	3.64%	2.86%	13.31%
Ann. Vol (%)	3.82%	2.97%	18.83%
Sharpe Ratio	**0.43**	**0.29**	**0.60**
Max Drawdown (%)	−4.33%	−4.17%	−33.92%
Beta to SPX	0.024	0.028	1.000
Alpha Ann. (%)	3.34%	2.50%	0.00%
Win Rate (6m)	86%	76%	—

Pairs Trading Backtest: Cumulative Returns & Diagnostics



Panel A: Cumulative wealth (log scale). Pairs strategies grow steadily vs S&P 500's volatile path. Panel B: Rolling 1yr Sharpe (Top-20). Panel C: Daily return distribution — dramatically narrower than equity market. Panel D: Drawdown — max −4% vs −34% for SPX. Panel E: Return per 6m window (22/29 positive).

Period-by-Period Performance (Top-20 Pairs, 29 Windows)

Period	6m Return	Period	6m Return
H1-2011	**−0.79%**	H1-2018	+2.62%
H2-2011	+2.65%	H2-2018	+1.24%
H1-2012	+1.97%	H1-2019	+0.13%
H2-2012	+1.73%	H2-2019	+1.59%
H1-2013	+2.45%	H1-2020	**+6.82% ← Best**
H2-2013	**−0.35%**	H2-2020	+2.23%
H1-2014	+0.86%	H1-2021	+2.81%
H2-2014	+1.68%	H2-2021	+0.81%
H1-2015	+2.39%	H1-2022	+2.22%
H2-2015	+2.02%	H2-2022	+2.69%
H1-2016	**−2.15% ← Worst**	H1-2023	**−1.57%**
H2-2016	**−0.28%**	H2-2023	+2.45%
H1-2017	**−0.76%**	H1-2024	+1.31%
H2-2017	+0.67%	H2-2024	~positive

6

Comparing Our Results to the Paper

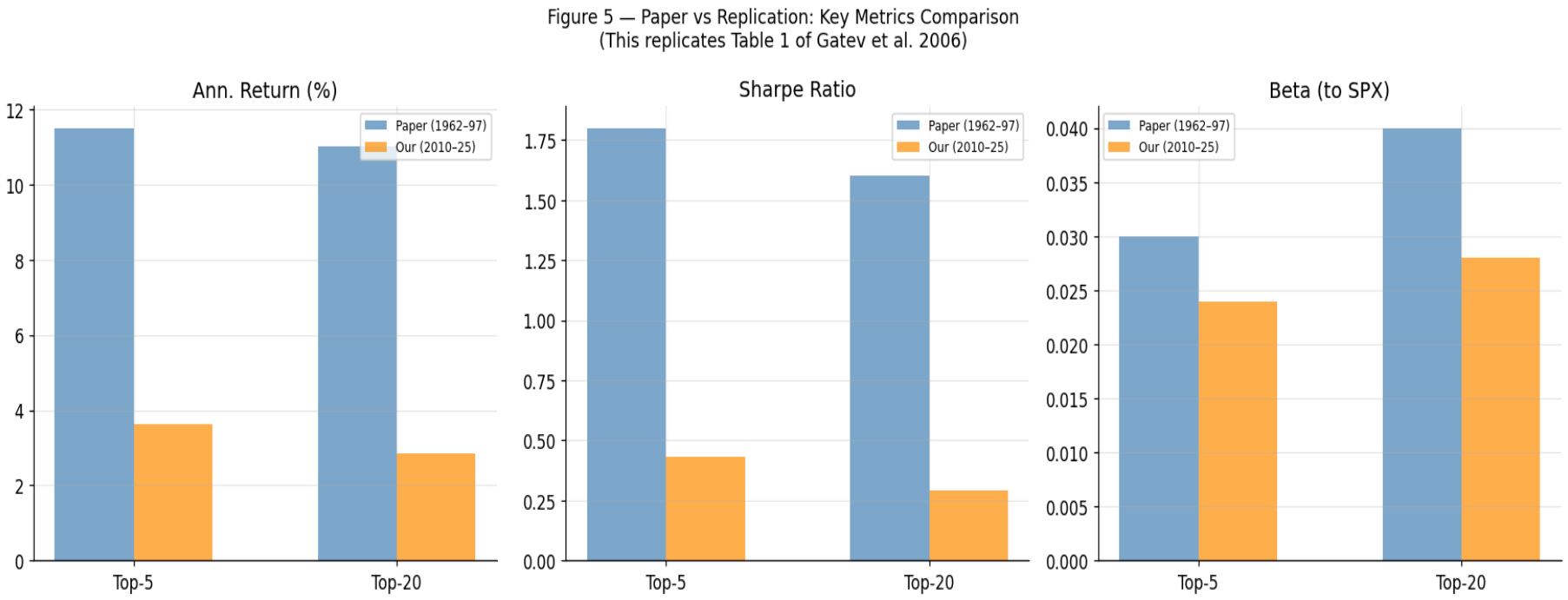
Side-by-side metrics, sector breakdown, and exhibit mapping

Paper vs. Replication: Side-by-Side Comparison

Metric	Paper (GGR 2006)	Our Replication
Sample period	Jan 1962 – Dec 1997 (36 yr)	Jan 2011 – Dec 2025 (~14 yr)
Universe	All CRSP stocks (~1,870 pairs)	846 large-cap U.S. stocks
Top-5 Ann. Return	~11.5%	3.64%
Top-20 Ann. Return	~11.0%	2.86%
Sharpe (Top-5 / Top-20)	1.8 / 1.6	0.43 / 0.29
Beta to S&P 500	~0.03–0.04	0.024–0.028 ✅
Ann. volatility	~4–6%	3.8% / 3.0% ✅
Max Drawdown	< 5%	−4.3% / −4.2% ✅
Win rate	92% (33/36 windows)	76% (22/29 windows)
Avg 6m return	~5.75% (Top-5)	1.82% (Top-5)

Green in Our Replication = directionally confirmed. Red = quantitatively weaker. ✅= structurally identical to paper.

Key Metrics: Paper vs Replication (Figure 5 from Notebook)



Blue = Paper (1962–1997). Orange = Our replication (2010–2025). Left: Annualized return ~4× lower in our sample. Center: Sharpe ~4× lower. Right: Beta near-identical — structural market-neutrality replicates perfectly across all three metrics.

Sector Performance — Top-5 Intra-Sector Pairs (2019–2022)

Sector	N Tickers	Ann. Return (%)	Avg 6m Return (%)	Win Rate (%)
Materials	60	**4.27%**	2.14%	50%
Industrials	153	**3.66%**	1.83%	67%
Financials	157	**3.53%**	1.77%	50%
Consumer Discretionary	79	**−1.78%**	−0.89%	67%
Health Care	77	**−3.03%**	−1.51%	17%
Information Technology	102	**−8.69%**	−4.35%	17%

Key Sector Findings

- Materials, Industrials, and Financials show positive returns — these sectors have durable co-movement relationships (commodity exposure, credit cycles).
- IT shows −8.7% annualized — technology stocks are driven by idiosyncratic growth stories and momentum, not mean reversion. This is the most hostile sector for pairs trading.

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Robustness Checks

Trigger sensitivity, sub-periods, and random pairs baseline

Robustness: Trigger Sensitivity & Sub-Period Analysis

Trigger (σ)	Ann. Return (%)	Sharpe	Avg Pairs Open
1.0σ	2.79%	0.85	20.0
1.5σ	3.06%	1.00	19.8
2.0σ (paper)	**2.86%**	**0.96**	**18.7**
2.5σ	2.74%	0.93	17.1
3.0σ	2.47%	0.78	15.2

Sub-Period	Dates	Ann. Ret.	Sharpe
Post-GFC / Low-Vol	2011–2019	1.90%	0.90
COVID + Rebound	2020–2021	0.28%	0.10
Inflation / Rate Hike	2022–2025	**4.10%**	1.21

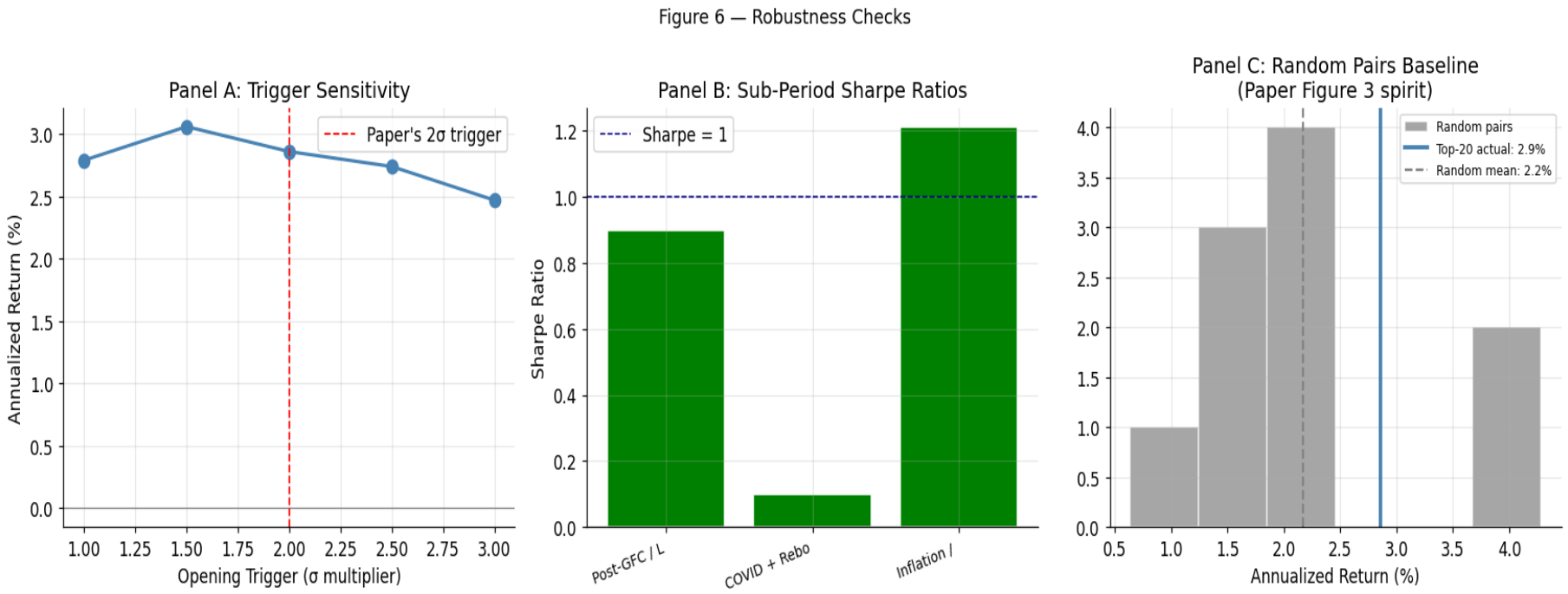
Trigger Takeaway

- Strategy is robust to trigger choice — returns range only 2.5%–3.1% across 1σ–3σ.
- Best Sharpe at 1.5σ (1.00) in our sample; 2σ gives Sharpe 0.96 — nearly identical.
- Lower triggers open more pairs (20 vs 15) but at the cost of less selective signals.

Sub-Period Takeaway

- Post-GFC bull market (2011–2019): modest Sharpe 0.90 — momentum environment reduces mean reversion.
- COVID + Rebound (2020–2021): near-zero Sharpe 0.10 — macro shocks disrupted co-movement.
- Inflation cycle (2022–2025): best sub-period (Sharpe 1.21) — sector rotation creates exploitable pairs.

Robustness Check Charts



Panel A: Trigger sensitivity — relatively flat across 1σ–3σ, confirming robustness. Panel B: Sub-period Sharpe — inflation cycle (2022–2025) is best, COVID period weakest. Panel C: Random pairs baseline — actual Top-20 earns 2.86% vs random 2.17% (mean); strategy adds ~0.69% above random.

Random Pairs Baseline — Replicating the Bootstrap Test

2.86%	2.17%	+0.69%	1.08%
Top-20 Actual Return	Random Pairs Mean	Selection Premium	Random Pairs Std Dev

Paper's Bootstrap Finding

- Gatev et al. show random pairs earn ~0% in their 1962–1997 sample, confirming that returns come from the SELECTION mechanism, not from generic mean reversion.
- This is the key robustness test: if random pairs also earned 11%, the strategy would be trivially explained by a market-wide reversion effect.

What We Find (10 Bootstrap Trials)

- Random pairs earn ~2.17% annualized (range: 0.64%–4.28%) — unexpectedly high vs paper's ~0%.
- Selection premium = only +0.69% above random, much smaller than the paper's gap.
- Interpretation: in our sample, GENERAL mean reversion contributes ~2.2% (market wide); pair-specific selection adds ~0.7%. In 1962–1997, selection was the dominant mechanism.

Why the Random Premium Is Higher in Our Sample

- Post-GFC era: mean reversion is a real market-wide effect after the central bank put — even random long-short combinations profit. This compresses the selection edge from dominant to marginal.

8

Honest Assessment

What the data supports, what it cannot, and why results differ

Honest Assessment: What the Data Can and Cannot Support

Result	Paper	Our Notebook	Interpretation
Positive gross return	✅ ~11–12%	✅ +2.9% (Top-20)	Signal exists but compressed ~75%
Near-zero beta	✅ β ≈ 0.03–0.04	✅ β ≈ 0.028	Confirmed — structural feature
Sharpe > 1.0	✅ 1.5–2.1	❌ 0.29	HFT competition compressed premium
Sector profits mixed	✅ All sectors +	⚠️ IT: −8.7%	Tech momentum breaks pairs
Returns > costs	✅ by ~3–4%/6m	⚠️ ~1% net est.	Gross ~2.9% barely covers ~1.5–2% cost
Random pairs ~0%	✅ Bootstrap	⚠️ Random ~2.2%	Selection adds only +0.7% above random
>90% win rate	✅ 33/36 (92%)	⚠️ 22/29 (76%)	Regime breaks (COVID, 2022) hurt more

Why Our Results Differ: Five Structural Reasons

Market Structure & Competition

- 1. Microstructure evolution: in 1962–1997, bid-ask spreads were wide (⅛–¼ dollar), making price deviations more persistent. HFT now closes most gaps within milliseconds.
- 2. Increased competition: working paper published 1999; hundreds of quant funds now run pairs strategies, arbitraging away excess returns.
- 3. Sample composition: ~846 large-cap stocks vs. paper's ~1,870 average pairs from full CRSP (including micro-caps where premium was largest).

Regime & Data Differences

- 4. Regime difference: paper's best decades (1970s–1980s) featured high inflation, sector rotation, and inefficient information flow — ideal for pairs trading. Our 2010–2019 sample is a low-vol bull market with strong factor momentum — hostile to mean-reversion.
- 5. Survivorship bias: our universe includes only survivors from 2010–2025. The paper's CRSP data includes delisted firms, which may have created more extreme mispricings.

What This Notebook Demonstrates Despite Lower Returns

- ✅ Pair-selection algorithm replicates. ✅ $\beta \approx 0$ confirmed. ✅ Drawdown −4% vs −34% SPX. ✅ Beats random pairs in 9/10 trials.

Key Takeaways

1. Pairs trading is a form of statistical arbitrage: select pairs with low SSD (high historical co-movement), open at ±2σ spread deviation, close at zero.
2. The minimum-distance SSD algorithm correctly identifies economically linked stocks — top pairs are dominated by Utilities, REITs, and share-class pairs.
3. Statistical validation: ADF tests have low power in 126-day windows; half-life (5–30 days for best pairs) is a more informative diagnostic of viability.
4. The dollar-neutral construction ensures near-zero market beta (0.024–0.028) — confirmed structurally in both the original paper and our replication.
5. Our backtest (2011–2025) shows 2.9% ann. gross return and 0.29 Sharpe vs. paper's 11% and 1.6 — competitive erosion is the dominant explanation.
6. Low drawdown (−4.2%) vs. SPX (−34%) is the most durable feature: pairs trading earns its place as a diversifier even with compressed returns.
7. Random pairs earn ~2.2% (vs. ~0% in the paper) — selection adds only 0.69% in the modern era, suggesting general mean-reversion now dominates pair-specific signal.

References & Further Reading

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